

Problem

Evaluate the following integrals:

$$\int_{-\infty}^{\infty} (t^3 + 5t^2 + 10)\delta(t+3) dt, \quad \int_0^{10} \delta(t-\pi) \cos 3t dt$$

Step-by-step solution

Step 1 of 2

(i) Given integral function is $\int_{-\infty}^{\infty} (t^3 + 5t^2 + 10)\delta(t+3) dt$

We know the shifting property

$$\int_a^b f(t)\delta(t-t_0) dt = f(t_0)$$

Therefore we can write

$$\int_{-\infty}^{\infty} (t^3 + 5t^2 + 10)\delta(t+3) dt = t^3 + 5t^2 + 10 \Big|_{t=-3}$$

$$\int_{-\infty}^{\infty} (t^3 + 5t^2 + 10)\delta(t+3) dt = (-3)^3 + 5(-3)^2 + 10$$

$$\int_{-\infty}^{\infty} (t^3 + 5t^2 + 10)\delta(t+3) dt = -27 + 45 + 10$$

$$\int_{-\infty}^{\infty} (t^3 + 5t^2 + 10)\delta(t+3) dt = 28$$

Step 2 of 2

(ii) Given integral function is $\int_0^{10} \delta(t-\pi) \cos 3t dt$

We know the shifting property

$$\int_a^b f(t)\delta(t-t_0) dt = f(t_0)$$

$$\int_0^{10} \delta(t-\pi) \cos 3t dt = \cos 3t \Big|_{t=\pi}$$

$$\int_0^{10} \delta(t-\pi) \cos 3t dt = \cos 3\pi$$

$$\int_0^{10} \delta(t-\pi) \cos 3t dt = -1$$